0 Start here

FULL

Jun's Garden: the guides

Quick version

No time to read it all? This is the short version. The full guide is below.

- These guides explain what the menu installed, how to look after your garden, and what to build next. The install itself is done by the menu (`00-READ-ME-FIRST.html`, one folder up).
- Rule 1: change things through the menu: Settings (menu `8`), users (menu `U`), parts (menu `9`), the front door (menu `M`, `1`). Back up (menu `6`) before any bigger change.
- Rule 2: passwords and keys live only in `.env`. Never put them in a chat, a screenshot, an email or an AI chat.
- Rule 3: for watering projects, use only low-voltage parts (5 V or 12 V). Never cut or wire anything that plugs into a wall outlet.
- The projects go from easy to hard. Start with Project 1 (Uptime Kuma, 20 minutes).
- AI is sometimes wrong in a confident voice. Check its command against the guide before you run it.
- Stuck? Run the checklist (menu `5`). Menu `H` makes a help file with no passwords in it.

This folder holds the guides for **your** garden website and for building your own home and garden projects. The install itself is done by the menu (see `00-READ-ME-FIRST.html` one

folder up). These guides explain what the menu did, how to look after the garden, and what to build next.

Everything here is made to be done one small step at a time. You can stop at any step and come back.

The 3 rules

1. **Change things through the menu.** Settings (menu `8`), users (menu `U`), parts (menu `9`) and the front door (menu `M`, `1`) keep everything in step. Back up (menu `6`) before any bigger change.
2. **Secrets stay secret.** Passwords, app passwords, the No-IP key and API keys live only in `.env` (the menu puts them there, hidden). They never go in a chat, a screenshot, an email, or a file you send to someone. This includes AI chats.
3. **Safe power only.** For watering projects, use low-voltage parts (5 V or 12 V) from a kit, and never cut or wire anything that plugs into a wall outlet. The projects say "STOP SIGN" where this matters.

What you have

Part	What it is
The website	Your plant catalogue, gallery, blog, data page, guestbook and plant admin, written in Rust and commented line by line
Garden tools (`/beds`)	Bed planner, checkup, diagnose, tag scan, spend, wiki queue (optional part)
Wiki	Your garden encyclopedia (optional part)
`garden.sh`	The menu that installs and looks after everything
`START-HERE/`	These guides

Part	What it is
`lab/`	Practice scripts that do single jobs, for learning (`lab/README.md`)
`docker/`	How each program is built, plus ready-made files for Uptime Kuma, Home Assistant, ESPHome and Apache
`AGENTS.md`, `GEMINI.md`	Rules that AI helpers read, so they help you safely
`WHAT-GETS-INSTALLED.md`, `WHY-THESE-PROGRAMS.md`	Every program, where it comes from, and why it was chosen

Searching the guides

The guides are many, so the handbook has a search box at the top of every page. Open `00-WELCOME.html` (in the garden folder; on Windows it opens after the install).

1. Click the search box at the top (or press `/`). With nothing typed, it shows common topics: click one, for example **router**, **backup** or **rename**.
2. Or type any word, for example `password` or `NAS`. Several words find the places that have all of them: `backup Google Drive`.
3. Next to the box, choose the language: **English + 한국어**, **English** or **한국어**. The menu remembers your choice.
4. Click a result: the guide opens at that very section. `Enter` opens the first result; `Esc` clears the box.

It searches every guide here, in both languages, and the other guides too (the n8n and project guides, what gets installed, the summary, the journal). It works on this computer, with no internet, and nothing you type is sent anywhere.

Looking after your garden

Guide	What it covers
03 Use your garden at home	Your password, email, the AI boxes, your first edit, the everyday menu
23 The full setup	Everything, the way the garden runs today, in order: choose your level (Easy, Recommended or Full), then one part at a time
11 How your website works	Where each file lives, and the safe way to change things
12 Your database	The tables, looking inside, safe habits
26 Rename, move, switch off and tidy up	Your own names for folders and pages, moving to a bigger disk, the on / off switch, Docker on Windows
13 Backups and folder names	Backup names, restoring, moving and renaming folders
14 Reserve your IP, open two ports	The router side
04 Put it on the internet	Your No-IP account, the front door, practising first
04b Moving junsgarden.zapto.org	What is already yours, and what to check
07 Your sign-in page	How the sign-in page and its gates work, guests, two-factor
09 Troubleshooting	Fixes by symptom
25 Common questions	Short answers with the menu keys: restarts, passwords, friends, undo, photos, updates, moving

Learning and AI

Guide	What it covers
16 Google AI Studio and Gemini	Searching the web with sources, coding help, learning about your computer, a free key

Guide	What it covers
08 Gemini setup	Gemini CLI on the laptop
05 Gemini prompts	Copy-and-paste starters
15 Local AI	AI on a strong home computer (optional, advanced)
17 Your Android app, and coding it with AI	Android Studio, the app on your phone, Gemini in Android Studio, local AI, the safe way to change code
00b Before you start	Everything you need, necessary and optional, with how long each part takes. Read this first
20 SSH and Remote Desktop	Work on the garden computer from your Windows PC; keys; copying files; seeing its screen
22 Bash, step by step	The terminal way, harder but not tough: a first walkthrough of the terminal, the prompt, your first commands, running the garden program
24 Tools for learning and programming	Very optional: editors, Python, Git and other tools for Windows, Ubuntu and the Raspberry Pi, and whether each one is already there
21 Every program and where it comes from	Each program, necessary and optional, its version, why, and its maker's own documents; how Let's Encrypt gives the padlock
19 Every menu option, explained	Each choice in every menu: what it does, when to use it, what it changes, how to undo it
18 Review the code yourself	A map of the files, checking nothing changed, git history, the secret scanner, asking Gemini or a friend to review it

Projects (the ladder, easy to hard)

Step	Guide	You will have	Time	Level
1	Project 1: Uptime Kuma	A dashboard that tells you if your site is up	20 min	easy

Step	Guide	You will have	Time	Level
2	Project 2: Backups	Copies of your backups off the laptop	30 min	easy
3	Project 3: Home Assistant	Your own smart-home hub	45 min	medium
4	Project 4: Govee lights	Lights that follow sunset and your rules	30 min	easy once step 3 works
5	Project 5: Weather and rain	Rain and frost alerts, and new weather readings	30 min	medium
6	Project 6: Soil sensor	A sensor that tells you when a bed is dry	2 h	medium
7	Project 7: Automatic watering	A valve or pump that waters by itself, safely	3 h	medium to hard
8	Project 8: Readings on your website	Your sensor numbers on your own Data page	3 h	hard
9	Project 9: More ideas	A list of next projects to pick from		choose

The n8n guides (email and automation) fit after Project 5. Switch n8n on first (menu `9`): n8n 1: First email, n8n 2: A dry-soil email, n8n 3: A weekly garden report.

If you set up a laptop from scratch: 01 Ubuntu on the laptop and 02 Docker explain what the install's first steps do. Curious what Docker is, without the technical part? 02a What is Docker? (two pictures, plain words). 10 The plan is a careful week-by-week order with checkpoints.

Also useful: the shopping list and your wins checklist.

How each guide is laid out

Every guide has the same parts in the same order, so you always know where you are:

1. **What you will build**
2. **Time and cost**
3. **What you need first**
4. **Steps** (numbered, with what you should see after each one)
5. **How you know it worked**
6. **If something goes differently**
7. **Make it yours** (optional changes to try)

Using Gemini as your helper

- **Give it the context first.** Open Gemini prompts, copy the "context block", and paste it as your first message in a new chat.
- **Paste the exact error.** Copy the red text from the terminal word for word.
- **Ask it to explain.** "Explain this command in plain words before I run it" is always a good question.

AI is sometimes wrong, in a confident voice. **Check the command against the guide before you run it.** If they disagree, trust the guide first, then ask Gemini why they differ. Never paste a password or key into any AI chat. If you do by accident, change it (Settings, menu `8`).

When you get stuck

1. Run the checklist (menu `5`). It says what to fix first.
2. Read the "If something goes differently" part of the guide.
3. Menu `H` makes a help file (no passwords in it). Paste it into Gemini with the "I am stuck" prompt, or show it to someone you trust.
4. Still stuck after 20 minutes? Stop, take a break, and write down what you tried. Fresh eyes fix most problems.

Being stuck is normal. It happens to professionals every day. Solving it is the actual skill.

Three names for one computer (people mix these up all the time)

Name	What it is	Example	How to see it (in a terminal on that computer)
User name	YOUR name on that computer: the account you log in with. Each person has their own. SSH asks for it: `ssh USERNAME@ADDRESS`.	`jun`	`whoami`
Host name (device name)	The COMPUTER's own name, for people to recognise it on the network. Not a login.	`garden-lab`	`hostname`
IP address	The computer's NUMBER on your home network. The router gives it; reserve it so it never changes (guide 14). Programs use this to find the computer.	`192.168.1.50`	`hostname -I` (the first number)

So to reach the garden laptop from another computer you type `ssh jun@192.168.1.50`: **user name** `@` **IP address**. The host name (`garden-lab`) only helps you recognise it.

Official beginner pages: Change the device name (host name), Find your IP address, Add a new user account, User management on Ubuntu.

Words you will see

Word	Plain meaning
Terminal	The window where you type commands
Command	One line you type in the terminal, then press Enter
`sudo`	"Run this as the administrator". It asks for your password

Word	Plain meaning
Container	A small, sealed box that holds one program and everything it needs
Docker	The tool that makes and runs containers
Compose	A file (`docker-compose.yml`) that lists the containers to start together
Image	The recipe a container is made from
Port	A numbered door on a computer. `8030` is the door your website uses
localhost	"This computer". Only you can see addresses that start with it
`.env`	The file holding your private settings. The menu changes it for you
Reverse proxy	A front door program (Caddy) that sends visitors to the right container
SSO / sign-in page	One login page in front of several tools (yours is Authelia)
DDNS	A service that keeps a web name pointed at your home internet address
HTTPS	The padlock in the browser. It means the connection is private

Have fun with it. You are building real things.